

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Module - II

(Backend Development Using Node.js, Express, and MongoDB)

Roll No.: T013	Name: Saqlain Khan
Class: TYIT	Batch: 1
Date of Assignment: 30-09-2026	Date/Time of Submission: 30-09-2026

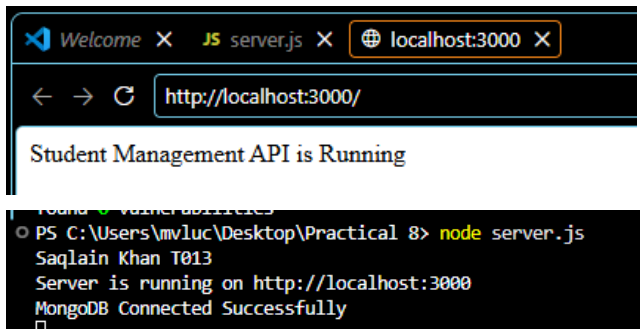
AIM :- API Creation and Integration

A) Create a simple REST API using Express

CODE :-

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
.then(() => {
  console.log("MongoDB Connected Successfully");
})
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

OUTPUT :-

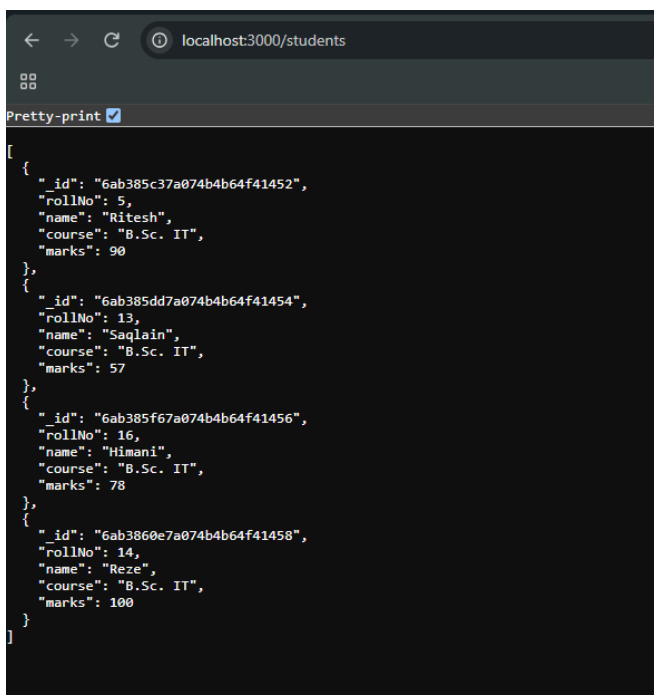


B) Implement a GET API to retrieve data.

CODE :-

```
Add this code above before app.listen()
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("students")
      .find()
      .toArray();
    res.json(students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
```

OUTPUT :-



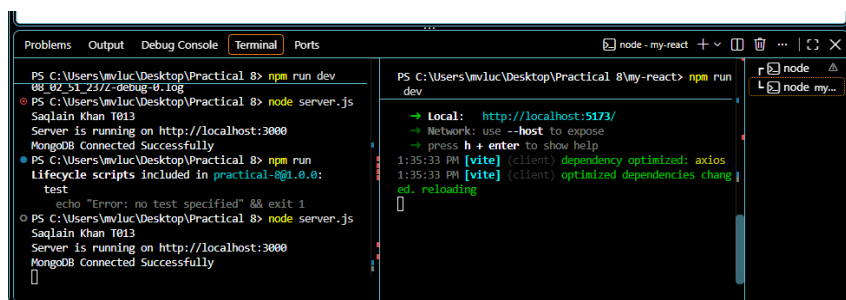
C) Fetch data from the API in a React application.

CODE :-

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  useEffect(() => { axios.get("http://localhost:3000/students")
    .then((response) => {setStudents(response.data);})
    .catch((error) => {console.log(error);}); }, []);
  return (
    <div>
      <h1>Student List</h1>
      <table border="1">
        <thead>
          <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Course</th>
            <th>Marks</th>
          </tr>
        </thead>
        <tbody>
          {students.map((student) => (
            <tr key={student._id}>
              <td>{student.rollNo}</td>
              <td>{student.name}</td>
              <td>{student.course}</td>
              <td>{student.marks}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}
```

export default App;

OUTPUT :-



Student List

Roll No	Name	Course	Marks
5	Ritesh	B.Sc. IT	90
13	Saqlain	B.Sc. IT	57
16	Himani	B.Sc. IT	78
14	Reze	B.Sc. IT	100

D) Send data from React to the server using a POST request.

CODE :-

App.jsx :-

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");
  // Get students
  const getStudents = () => {
    axios.get("http://localhost:3000/students")
      .then((response) => {
        setStudents(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  };
  useEffect(() => {
    getStudents();
  }, []);

  // Add student
  const addStudent = () => {
    const newStudent = {
      rollNo: Number(rollNo),
      name: name,
      course: course,
      marks: Number(marks)
    };
    axios.post("http://localhost:3000/students", newStudent)
      .then((response) => {
        alert(response.data.message);
        setRollNo("");
        setName("");
        setCourse("");
        setMarks("");
        getStudents();
      })
  }
```

```
.catch((error) => {
console.log(error);
});
};
return (
<div>
<h1>Student Management System</h1>
<h2>Add Student</h2>
<input
type="number"
placeholder="Roll No"
value={rollNo}
onChange={(e) => setRollNo(e.target.value)}
/>
<br /><br />
<input
type="text"
placeholder="Student Name"
value={name}
onChange={(e) => setName(e.target.value)}
/>
<br /><br />

<input
type="text"
placeholder="Course"
value={course}
onChange={(e) => setCourse(e.target.value)}
/>
<br /><br />
<input
type="number"
placeholder="Marks"
value={marks}
onChange={(e) => setMarks(e.target.value)}
/>
<br /><br />
<button onClick={addStudent}>
Add Student
</button>
<hr />
<h2>Student List</h2>
<table border="1">

```

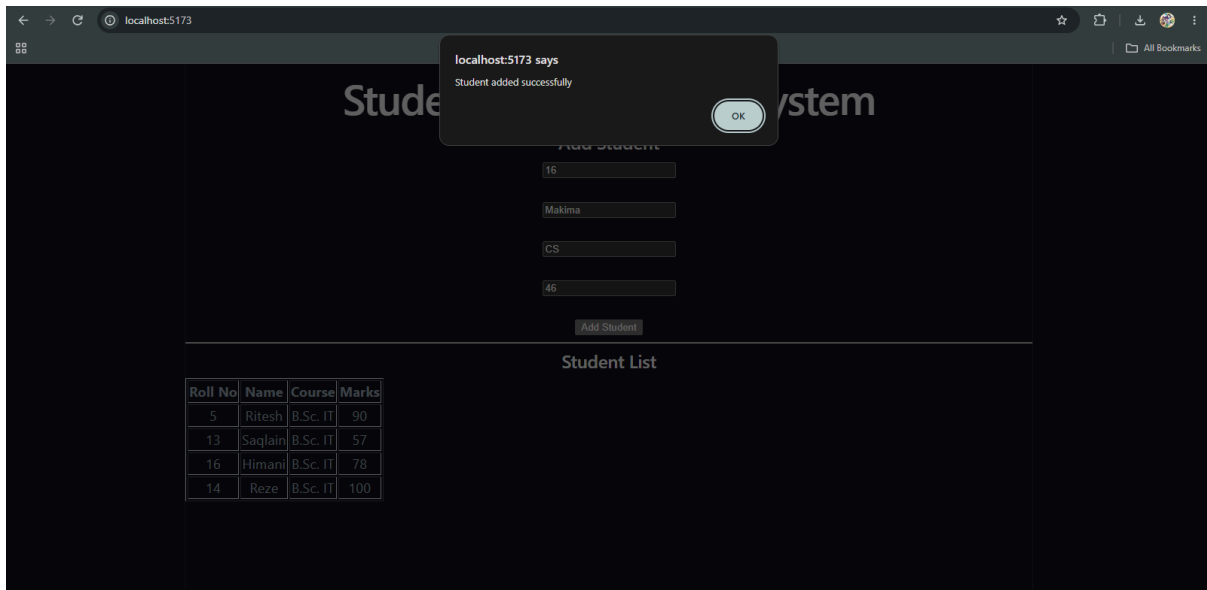
```

<tr>
<th>Roll No</th>
<th>Name</th>
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</tr>
</thead>
<tbody>
{students.map((student) => (
<tr key={student._id}>
<td>{student.rollNo}</td>
<td>{student.name}</td>
<td>{student.course}</td>
<td>{student.marks}</td>
</tr>
))}
</tbody>
</table>

</div>
);
}
export default App;
server.js (Add it above app.listen()) :-
app.post("/students", async (req, res) => {
try {
const student = req.body;
await mongoose.connection.db
.collection("students")
.insertOne(student);
res.json({
message: "Student added successfully",
student: student
});
}
catch (error) {
res.status(500).json({
message: "Error adding student"
});
}
});

```

OUTPUT :-



Roll No	Name	Course	Marks
5	Ritesh	B.Sc. IT	90
13	Saqlain	B.Sc. IT	57
16	Himani	B.Sc. IT	78
14	Reze	B.Sc. IT	100
16	Makima	CS	46

E) Store data in MongoDB.

F) Display stored data in React page.

CODE :-

App.jsx :-

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
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  const [marks, setMarks] = useState("");
  // Get students
  const getStudents = () => {
    axios.get("http://localhost:3000/students")
      .then((response) => {
        setStudents(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  };
}
```

```

};
useEffect(() => {getStudents();}, []);
// Add student
const addStudent = () => {const newStudent = {
rollNo: Number(rollNo),
name: name,
course: course,
marks: Number(marks)
};
axios.post("http://localhost:3000/students", newStudent)
.then((response) => {alert(response.data.message);
setRollNo("");
setName("");
setCourse("");
setMarks("");
getStudents();
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.catch((error) => {
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<br /><br />
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placeholder="Student Name"
value={name}
onChange={(e) => setName(e.target.value)}
/>
<br /><br />
<input
type="text"
placeholder="Course"
value={course}
onChange={(e) => setCourse(e.target.value)}

```

```

/>
<br /><br />
<input
type="number"
placeholder="Marks"
value={marks}
onChange={(e) => setMarks(e.target.value)}
/>
<br /><br />
<button onClick={addStudent}>
Add Student
</button>
<hr />
<h2>Student List</h2>
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<td>{student.name}</td>
<td>{student.course}</td>
<td>{student.marks}</td>
</tr>
))}
</tbody>
</table>
</div>
);
}
export default App;

```

server.js :-

```

const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

```

```

const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
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.then(() => {
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  try {
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      .toArray();
    res.json(students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
app.post("/students", async (req, res) => {
  try {
    const student = req.body;
    await mongoose.connection.db
      .collection("students")
      .insertOne(student);
    res.json({
      message: "Student added successfully",
      student: student
    });
  }
  catch (error) {

```

```
res.status(500).json({
message: "Error adding student"
});
}
});
app.listen(PORT, () => {
console.log(` Server is running on http://localhost:${PORT}`);
});
```

OUTPUT :-

Student Name

Course

Marks

Add Student

Student List

Roll No	Name	Course	Marks
13		B.Sc. IT	
5		B.Sc. IT	
16		B.Sc. IT	
14		B.Sc. IT	
16		CS	
13	Saqlain Khan	TYIT	78
14	Reze	CS	35